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T12 – AI&DS

SUB : ASD&D

EXPERIMENT – 2

AIM : To Containerize Application using Docker.

LO Mapped : LO4

LO4 : Containerize , orchestrate , and deploy applications using Docker , Docker Compose , Kubernetes and Cloud Infrastructure .

THEORY :

1. Introduction to Docker

Docker is an open-source platform used to develop, package, deploy, and run applications in a standardized environment called a container. It uses operating-system-level virtualization to isolate applications and their dependencies without requiring a complete virtual machine.

Docker solves the common problem of "it works on my machine" by packaging an application along with its required libraries, dependencies, and configuration. This allows the application to run consistently across different environments.

2. Containerization

Containerization is the process of packaging an application and all its required dependencies into a lightweight, portable container.

Containers share the host operating system's kernel, making them generally faster and more lightweight than traditional virtual machines.

Advantages of containerization include:

- Portability

- Fast startup time
- Lightweight resource usage
- Application isolation
- Easy deployment
- Consistent environments
- Scalability

3. Docker Image

A Docker image is a read-only, immutable template that contains everything required to run an application. It includes:

- Application source code
- Runtime environment
- Libraries
- Dependencies
- Environment configuration
- Instructions for starting the application

Images are commonly created using a Dockerfile. An image can be used to create multiple containers.

For this experiment, the Docker image was named:

My-python-app

4. Docker Container

A Docker container is a running or stopped instance of a Docker image. It provides an isolated environment where the application executes.

A container is created from an image using the docker run command. Multiple containers can be created from the same image, and each container can have its own name, network configuration, and runtime state.

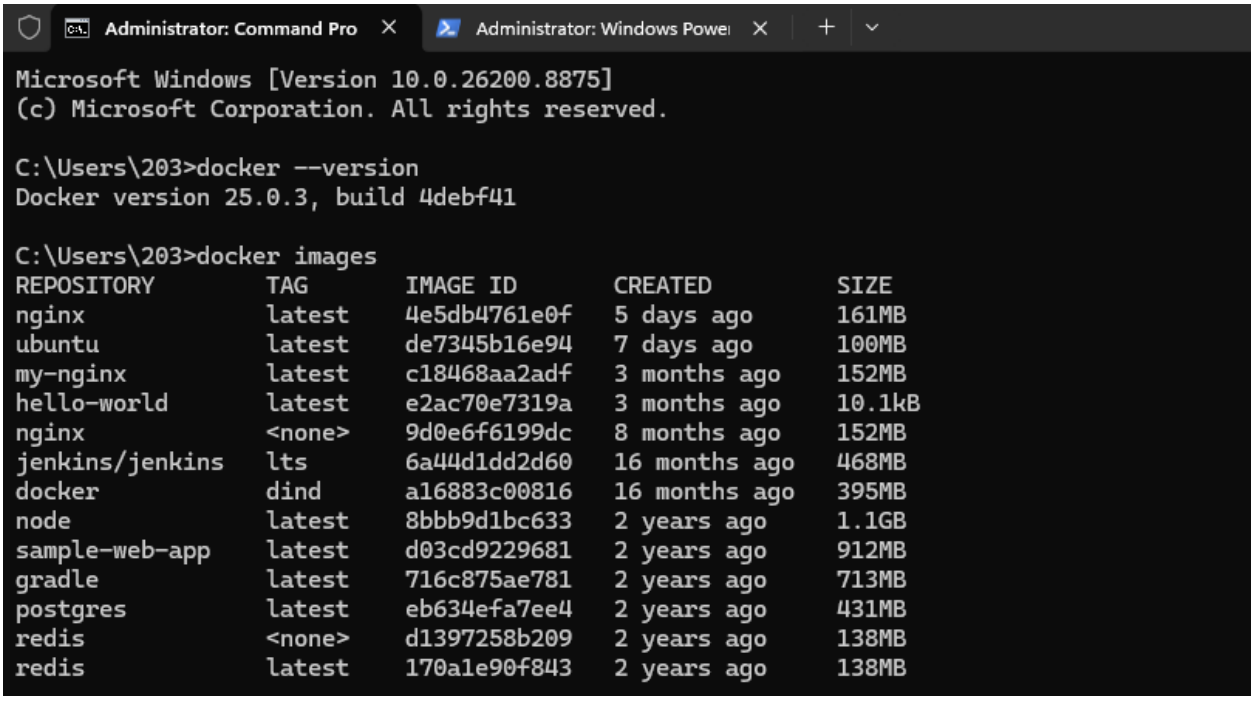
In this experiment, the container was named:

My-python-app-container

5. Difference Between Docker Image and Container

Feature	Docker Image	Docker Container
Definition	Template used to create containers	Instance of an image
State	Immutable/read-only	Has a runtime state
Purpose	Stores application and dependencies	Runs the application
Creation	Built using a Dockerfile	Created using docker run
Example	My-python-app	My-python-app-container
Multiple Instances	One image can create many containers	Each container is an individual instance
Lifecycle	Exists until deleted	Can be started, stopped, and removed

OUTPUT :



```
C:\Users\203>docker search ubuntu
NAME                            DESCRIPTION                                     STARS     OFFICIAL
ubuntu                          Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid                    Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx                    Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9                    BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/kafka                    Apache Kafka, a distributed event streaming …   62
ubuntu/zookeeper                ZooKeeper maintains configuration informatio…   15
ubuntu/apache2                  Apache, a secure & extensible open-source HT…   104
ubuntu/prometheus               Prometheus is a systems and service monitori…   80
ubuntu/jre                      Chiseled Java runtime based on Ubuntu. Long-…   25
ubuntu/dotnet-aspnet            Chiselled Ubuntu runtime image for ASP.NET a…   26
ubuntu/python                   A chiselled Ubuntu rock with the Python runt…   33
ubuntu/mysql                    MySQL open source fast, stable, multi-thread…   74
ubuntu/postgres                 PostgreSQL is an open source object-relation…   45
ubuntu/dotnet-runtime           Chiselled Ubuntu runtime image for .NET apps…   24
ubuntu/redis                    Redis, an open source key-value store. Long-…   26
ubuntu/dotnet-deps              Chiselled Ubuntu for self-contained .NET & A…   16
ubuntu/grafana                  Grafana, a feature rich metrics dashboard & …   12
ubuntu/memcached                Memcached, in-memory keyvalue store for smal…   7
ubuntu/jdk                      Ubuntu rock for OpenJDK                         0
ubuntu/rabbitmq                 Ubuntu-based RabbitMQ image - an open source…   0
ubuntu/mimir                    Ubuntu ROCK for Mimir, a horizontally scalab…   0
ubuntu/loki                     Grafana Loki, a log aggregation system like …   2
ubuntu/traefik                  Ubuntu ROCK for Traefik, a modern HTTP rever…   3
ubuntu/opentelemetry-collector  This is a rock for the OpenTelemetry Collect…   0
ubuntu/alertmanager             Ubuntu Rock for Alertmanager. It handles ale…   1

C:\Users\203>docker search ubuntu --filter STARS=100
NAME             DESCRIPTION                                     STARS     OFFICIAL
ubuntu           Ubuntu is a Debian-based Linux operating sys…   17859     [OK]
ubuntu/squid     Squid is a caching proxy for the Web. Long-t…   129
ubuntu/nginx     Nginx, a high-performance reverse proxy & we…   141
ubuntu/bind9     BIND 9 is a very flexible, full-featured DNS…   120
ubuntu/apache2   Apache, a secure & extensible open-source HT…   104
```

```
C:\Users\203>docker run ubuntu /bin/bash

C:\Users\203>docker ps -a
CONTAINER ID   IMAGE                 COMMAND                  CREATED          STATUS                      PORTS                                NAMES
79d59be7b7ef   ubuntu                "/bin/bash"              10 seconds ago   Exited (0) 8 seconds ago                                         quirky_matsumoto
c8fe1406dd791   ubuntu                "bash"                   19 hours ago     Exited (255) 7 minutes ago                                       pedantic_hypatia
2ba1c97abb52   ubuntu                "/bin/bash"              19 hours ago     Exited (127) 19 hours ago                                        suspicious_hugle
0fb51210958c   my-nginx              "/docker-entrypoint.…"   3 months ago     Exited (255) 19 hours ago   0.0.0.0:9091->80/tcp                 thirsty_pare
fe399986e504   9d0e6f6199dc          "/docker-entrypoint.…"   3 months ago     Exited (255) 19 hours ago   0.0.0.0:9090->80/tcp                 infallible_germain
5be8ea3f3bb2   9d0e6f6199dc          "/docker-entrypoint.…"   3 months ago     Created                                                          ecstatic_jennings
1e61b2cb20bf   hello-world           "/hello"                 3 months ago     Exited (0) 3 months ago                                          nervous_elbakyan
23106efee7f4   9d0e6f6199dc          "/docker-entrypoint.…"   8 months ago     Exited (255) 4 months ago   0.0.0.0:8081->80/tcp                 my-nginx
d1c21cfdc92b   jenkins/jenkins:lts   "/usr/bin/tini -- /u…"   16 months ago    Exited (255) 8 months ago   50000/tcp, 0.0.0.0:9090->8080/tcp    jenkins
6d895ae2b5eb   d1397258b209          "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           nice_greider
1f99a9cc77bf   gradle                "/__cacert_entrypoin…"   2 years ago      Exited (0) 2 years ago                                           elastic_stonebraker
77e3fc5c920f   node                  "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           con234
7b58e9788533   node                  "docker-entrypoint.s…"   2 years ago      Exited (0) 2 years ago                                           containerblah
0dc3412adadd   postgres              "docker-entrypoint.s…"   2 years ago      Exited (1) 2 years ago                                           stoic_mclaren

C:\Users\203>docker run -it ubuntu /bin/bash
root@9b4336400960c:/# mkdir demo
root@9b4336400960c:/# cd demo
root@9b4336400960c:/demo# pwd
/demo
root@9b4336400960c:/demo#
exit
```

```
C:\Users\Lab203\docker-app>docker run -it myubuntu /bin/bash
root@a700eba76fe1:/# mkdir docker-app
root@a700eba76fe1:/# cd docker-app
root@a700eba76fe1:/docker-app# apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [173 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [376 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [307 kB]
Error: Release file for http://archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease is not valid yet (invalid for another 1h 8min 16s). Updates for this repository will not be applied.
```

```
root@a700eba76fe1:/docker-app# apt install -y nano
Installing:
  nano

Suggested packages:
  hunspell

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 34
  Download size: 289 kB
  Space needed: 893 kB / 1022 GB available

Get:1 http://security.ubuntu.com/ubuntu resolute-security/main amd64 nano amd64 8.7.1-1ubuntu0.1 [289 kB]
Fetched 289 kB in 1s (281 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 1.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 1.)
debconf: falling back to frontend: Teletype
Selecting previously unselected package nano.
(Reading database ... 26589 files and directories currently installed.)
Preparing to unpack .../nano_8.7.1-1ubuntu0.1_amd64.deb ...
Unpacking nano (8.7.1-1ubuntu0.1) ...
Setting up nano (8.7.1-1ubuntu0.1) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/editor.1.gz because associated file /usr/share/man/man1/nano.1.gz (of link group editor) does not exist
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/pico.1.gz because associated file /usr/share/man/man1/nano.1.gz (of link group pico) does not exist
```

```
root@a700eba76fe1:/docker-app# nano app.py
root@a700eba76fe1:/docker-app# nano requirements.txt
root@a700eba76fe1:/docker-app# nano dockerfile
root@a700eba76fe1:/docker-app# ls
app.py  dockerfile  requirements.txt
root@a700eba76fe1:/docker-app# cat dockerfile
FROM python:3.12-slim

WORKDIR /app

COPY . .

CMD ["python", "app.py"]
root@a700eba76fe1:/docker-app# docker build -t my-python-app .
bash: docker: command not found
root@a700eba76fe1:/docker-app# exit
exit

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
```

```
C:\Users\Lab203\docker-app>docker cp
docker: 'docker cp' requires 2 arguments

Usage:  docker cp [OPTIONS] CONTAINER:SRC_PATH DEST_PATH|-
        docker cp [OPTIONS] SRC_PATH|- CONTAINER:DEST_PATH

Run 'docker cp --help' for more information

C:\Users\Lab203\docker-app>docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED          STATUS                      PORTS     NAMES
a700eba76fe1   myubuntu      "/bin/bash"              7 minutes ago    Exited (127) 2 minutes ago            friendly_rhodes
368f82d23c91   myubuntu      "/bin/bash"              20 minutes ago   Exited (0) 11 minutes ago             nostalgic_snyder
4f79701bdf0a   ubuntu        "/bin/bash"              29 minutes ago   Exited (127) 23 minutes ago           peaceful_ishizaka
30b568016772   ubuntu        "/bin/bash"              56 minutes ago   Up 56 minutes                         cool_meitner
50afc5d495c1   ubuntu        "/bin/bash"              56 minutes ago   Exited (0) 56 minutes ago             cranky_germain
c8d7d78063ff   nginx         "/docker-entrypoint.…"   19 hours ago     Created                               mynginx
db1e61f5826e   ubuntu        "bash"                   19 hours ago     Exited (127) 19 hours ago             friendly_buck
9252b211eddb   ubuntu        "/bin/bash"              20 hours ago     Exited (2) 20 hours ago               hungry_dhawan
1ec237a42c13   ubuntu        "/bin/bash"              20 hours ago     Exited (0) 20 hours ago               admiring_villani
b7d7f64bf20c   ubuntu        "/bin/bash"              20 hours ago     Exited (127) 20 hours ago             suspicious_bhaskara
96d324b8c9b3   ubuntu        "/bin /bash"             20 hours ago     Created                               ecstatic_ride
7d8611fb4d31   ubuntu        "/bin /bash"             20 hours ago     Created                               interesting_dijkstra
a5f07e90484b   ubuntu        "/bin/bash"              20 hours ago     Exited (0) 20 hours ago               sad_hofstadter
f8662871de4c   hello-world   "/hello"                 4 months ago     Exited (0) 4 months ago               hungry_bohr
cbdae8d0d2f5   hello-world   "/hello"                 4 months ago     Exited (0) 4 months ago               awesome_galileo

C:\Users\Lab203\docker-app>docker cp a700eba76fe1:/docker-app
docker: 'docker cp' requires 2 arguments

Usage:  docker cp [OPTIONS] CONTAINER:SRC_PATH DEST_PATH|-
        docker cp [OPTIONS] SRC_PATH|- CONTAINER:DEST_PATH

Run 'docker cp --help' for more information

C:\Users\Lab203\docker-app>docker cp a700eba76fe1:/docker-app C:\Users\Lab203
Successfully copied 4.1kB to C:\Users\Lab203
```

```
C:\Users\Lab203>cd docker-app

C:\Users\Lab203\docker-app>nano app.py
'nano' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Lab203\docker-app>notepad app.py

C:\Users\Lab203\docker-app>notepad requirements.txt

C:\Users\Lab203\docker-app>notepad dockerfile

C:\Users\Lab203\docker-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 3EB9-B80F

 Directory of C:\Users\Lab203\docker-app

21-07-2026  05:45    <DIR>          .
21-07-2026  05:38    <DIR>          ..
21-07-2026  05:54                29 app.py
21-07-2026  05:45                73 dockerfile
21-07-2026  05:43                 0 requirements.txt
               3 File(s)            102 bytes
               2 Dir(s)  742,369,075,200 bytes free
```

```
C:\Users\Lab203\docker-app>cd\

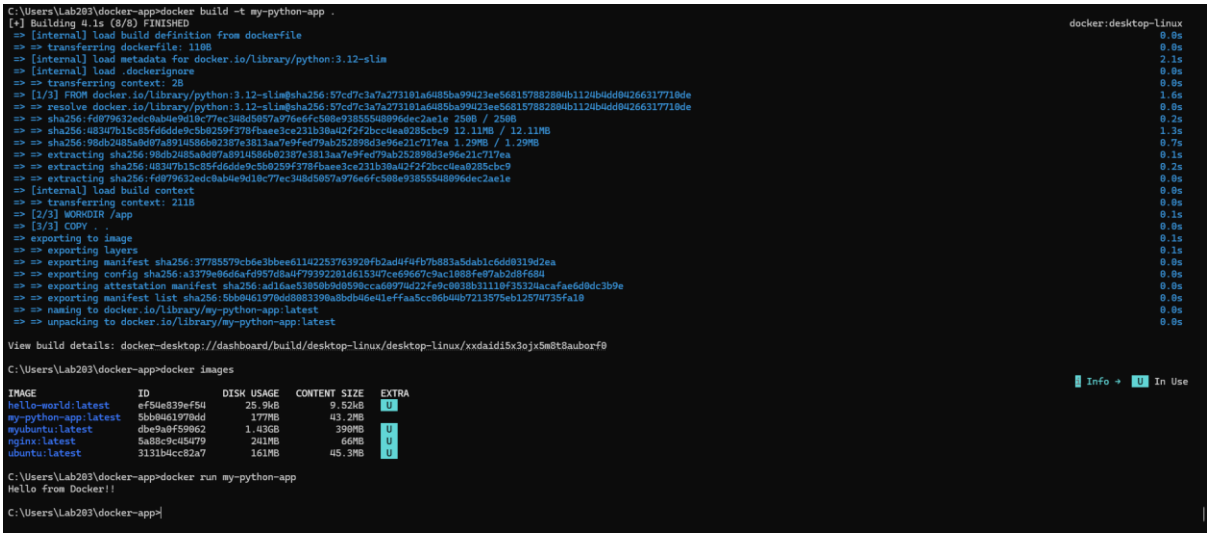
C:\>cd users

C:\Users>cd lab203

C:\Users\Lab203>dir
 Volume in drive C is Windows
 Volume Serial Number is 3EB9-B80F

 Directory of C:\Users\Lab203
```





CONCLUSION :

The experiment successfully demonstrated the basic use of Docker for containerizing and running a Python Flask application. We created a Docker image using a Dockerfile, ran and managed containers, viewed logs, inspected containers, and practiced essential Docker commands for images, containers, and Docker Hub.